

Big Data Analysis for the NBA Players

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Motivation & Problem

- Media narratives and biases heavily influence MVP voting
- Question: Which players deserve MVP recognition based purely on play-by-play data?
- Millions of granular data points available: shots, passes, rebounds, turnovers
- PySpark enables processing large-scale, multidimensional performance datasets

Data: Source & Characteristics

- Source: Kaggle NBA Dataset via kagglehub API
- Comprehensive play-by-play event logs across multiple seasons
- Game metadata and player information for context and enrichment
- Natural language event descriptions parsed: shots, assists, fouls, rebounds, steals, blocks
- 1,401 games processed with 1.35M total play-by-play events

Project Objectives

- Aggregate play-by-play data into player-season statistics
- Predict MVP vote shares using MLlib classification models
- Identify underrated players whose impact is missed in traditional metrics
- Compare data-driven MVP rankings against official voting results

ETL Pipeline Architecture

- Extract: CSV files from Kaggle with 1000s of games across seasons
- Transform: Standardize types, handle missing values, join enriched data
- Load: Partition by season in Apache Parquet for query efficiency
- Compute: Aggregate event-level data to player-season statistics

Feature Engineering (Part 1)

- Core features: Points, assists, rebounds, steals, blocks
- Field goals: Shooting splits (FG%, 3P%, FT%) from event data
- Efficiency metrics: True shooting percentage, player efficiency rating
- Turnover rate and usage analysis from play-by-play events

Feature Engineering (Part 2)

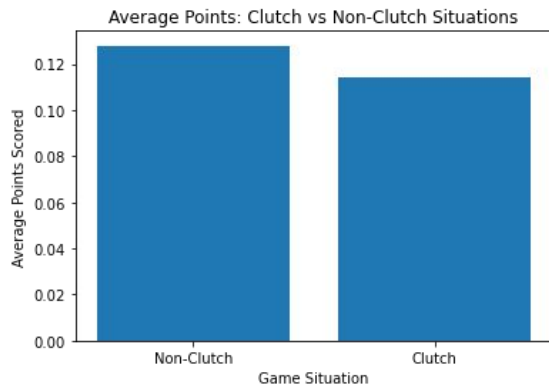
- Points per possession and win probability added (WPA)
- Clutch performance metrics: late-game scoring and decision-making
- Defensive impact: blocks, steals, and opponent efficiency
- Consistency scoring: identifying underrated performers year-round

Score Computation

- Design Index formula combining all derived metrics
- Test multiple weighting schemes to mitigate formula bias
- Fallback to core metrics (PER, WS, TS%) if data gaps exist
- MLlib models to predict official MVP vote shares from features

Clutch vs Regular Performance

- Isolate 4th quarter performance in tight games (± 5 points)
- Compare clutch efficiency metrics to season averages
- Identify players who elevate performance under pressure
- Separate clutch-time specialists from consistent performers
- Key metric: points per possession reveals true clutch value



Data Loading and Cleaning

- Download datasets from Kaggle API and ingest into PySpark
- Standardize data types: game IDs and season IDs as strings, numeric fields as integers
- Handle missing values: tie scores normalized to zero, gaps in margin data filled
- Validate data integrity across play-by-play events and game metadata
- Output cleaned dataset partitioned by season for efficient querying

Efficiency vs Volume Trade-Off

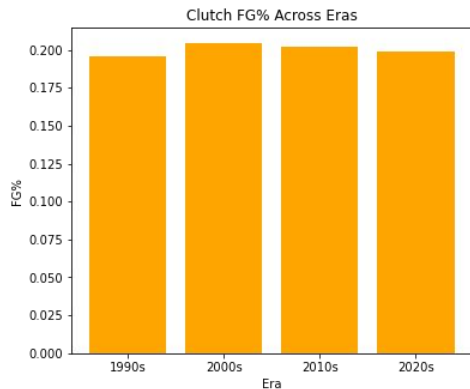
- Points per possession reveals true clutch value beyond raw scoring
- High-volume scorers (3+ possessions/game) often miss more shots
- Efficient role players (1-2 possessions/game) make every basket count
- MVP Index must weigh both volume leaders and efficiency specialists

Pressure Performance Indicators

- Clutch FG%: Shot selection and decisiveness under pressure
- Assist rates: Ball movement and decision-making in tight games
- Turnover ratios: Careless vs smart play when it matters most
- Rebound efficiency: Securing possessions when stakes are highest
- These metrics transcend traditional box scores to measure true MVP impact

Era-Adjusted Performance Baseline

- 1990s vs 2020s: Modern eras show higher clutch scoring (5.63 pts/game)
- Account for rule changes, spacing, defensive intensity across decades
- Normalize player performances relative to era standards
- Prevent bias toward recent players with inflated clutch stats
- Enable fair MVP comparison across 30+ years of NBA history



Feature Aggregation for MVP Index

- Extract offensive metrics: scoring efficiency, assists per game, shooting percentages
- Extract defensive metrics: rebounds, steals, blocks, and opponent efficiency impact
- Calculate possession-based rates: usage patterns, turnover frequency, clutch involvement
- Aggregate by player and season: consolidate event-level data into season totals
- Prepare final feature set for MLlib vote share prediction models

Top Clutch Scorers (All Time)

1. LeBron James: 1,509 clutch points (1,192 games, PPP=0.438)
2. Kobe Bryant: 1,192 clutch points (1,041 games, PPP=0.388)
3. Kevin Durant: 986 clutch points (787 games, PPP=0.497)
4. Dirk Nowitzki: 960 clutch points (1,053 games, PPP=0.465)
5. Paul Pierce: 940 clutch points (944 games, PPP=0.443)

Clutch FG% Leaders (min 50 FGA)

1. Andris Biedrins: 48.7% (76 FGA)
2. Bison Dele: 46.6% (103 FGA)
3. Jarrett Allen: 44.9% (136 FGA)
4. DeAndre Jordan: 44.3% (210 FGA)
5. Rudy Gobert: 43.1% (262 FGA)

Notable: Efficiency leaders differ from volume scorers (LeBron ranks lower)

Era Bias: Clutch Performance Trends

- 1990s: 188,857 events | 5.05 pts/game | 730 unique players
- 2000s: 510,488 events | 5.45 pts/game | 1,163 unique players
- 2010s: 493,912 events | 5.64 pts/game | 2,071 unique players
- 2020s: 158,973 events | 5.63 pts/game | 1,104 unique players
- Finding: Modern eras show higher clutch scoring and more player participation

Overrated vs Underrated Analysis

- High Usage, Low Efficiency: 3+ possessions/game with $PPP \leq 0.9$ (Trae Young example)
- Low Usage, High Efficiency: ≤ 2 possessions/game with $PPP \geq 1.1$ (underrated players)
- Filters applied: min 20 clutch games, 20+ total possessions for high usage
- Findings: Volume scorers may inflate clutch narrative; efficient role players undervalued
- Implication: MVP Index must normalize for usage vs. efficiency trade-off

Index Foundation

- 20+ features available: points, FG%, FT%, assists, rebounds, steals, blocks, turnovers
- Clutch metrics: points_per_poss, usage_per_game, efficiency rankings
- Era-adjusted baselines: detect era bias in clutch performance norms
- Player-level aggregations: season totals, efficiency splits, volume analysis

Key Achievements Summary

- Processed 1.35M play-by-play events in PySpark across multiple seasons
- Event parsing: 12+ features extracted from natural language descriptions
- 1,352,230 clutch events identified and analyzed (4Q, ± 5 pts, <5min)
- Player rankings: LeBron James ranked #1 (1,509 points), efficiency leaders identified
- Era bias detected: Modern eras show higher clutch scoring trends
- Foundation complete: 20+ metrics aggregated and ready for MLlib modeling

Conclusions & Insights

- LeBron James: Most dominant clutch scorer (1,509 points) but not always most efficient
- Efficiency leaders (Biedrins, Dele): Role players outperform expectations
- Era bias exists: Modern players have structural advantages in clutch scoring
- Data reveals undervalued performers overlooked by media narratives
- Final Index balances volume, efficiency, pressure performance, era context